

PROPOSAL FOR DISCUSSION

Shilpa Shakti Digital Stack

Six technology initiatives to speed up investment, jobs and ease of doing business in West Bengal

Prepared for: **Department of Industry, Commerce & Enterprises, Government of West Bengal**

Prepared by: **[Company Name]** · Kolkata · September 2026

THE SIX INITIATIVES

01

Nivesh Setu · Investor Lifecycle & MoU-to-Ground Tracker

Every rupee promised to Bengal, tracked until it becomes a factory and a payroll.

02

Shilpa Bhumi · Industrial Land Intelligence & Closed-Factory Revival Grid

Know every industrial acre: who owns it, whether it is clean, and whether it is being used.

03

Silpa Sathi 2.0 · Smart Clearance Engine: Rules-as-Code + Deemed-Approval Clock

Approvals in days, not months, with every decision explainable and every officer protected.

04

Karma Praman · Jobs-Linked Incentive Verification & Disbursal Engine

Pay incentives only for real jobs: verified, automatic and leak-proof.

05

Bangla Cluster Connect · MSME Cluster Commerce & Proof-of-Origin Network

From Bengal's looms, tea gardens and tanneries straight to buyers, with proof of origin.

06

Sabuj Shilpa · Green Industry DMRV & Carbon-Readiness Platform

Keep Bengal's industry compliant and export-ready, and let it earn from going green.

Each initiative is set out on two pages: problem, solution, approach, benefits and indicative cost. All figures are indicative and subject to detailed scoping with the Department.

Portfolio at a glance

West Bengal's new government has made industrial revival its first economic priority. Its plans include a jobs-linked incentive pool of ₹5,000 crore, a new land policy, an overhaul of Silpa Sathi, revival of closed-factory land, and a ₹1,000 crore programme to rejuvenate legacy clusters. Each of these commitments now needs a digital backbone that makes it fast, verifiable and visible to investors. The six initiatives below were chosen because they either do not exist yet or exist only as static portals, and because each one directly supports a commitment the government has already announced.

#	Initiative	What it solves	Status today	One-time	Recurring / yr
01	Nivesh Setu Investor Lifecycle & MoU-to-Ground Tracker	MoUs that never become factories; no single view of stuck projects	Not started as an integrated system	₹1.15–1.9 cr	₹35–60 lakh
02	Shilpa Bhumi Industrial Land Intelligence & Closed-Factory Revival Grid	Scarce, opaque and idle industrial land; closed-factory tracts	Partially started (static data only)	₹2.15–3.55 cr	₹75 lakh–1.3 cr
03	Silpa Sathi 2.0 Smart Clearance Engine: Rules-as-Code + Deemed-Approval Clock	Desk-to-desk approvals; 'deemed approval' with no enforcement	Partially started: overhaul underway	₹2.45–3.9 cr	₹75 lakh–1.2 cr
04	Karma Praman Jobs-Linked Incentive Verification & Disbursal Engine	Jobs-linked incentives that are easy to inflate; legacy-claim disputes	Not started: greenfield	₹1.8–2.75 cr	₹50–85 lakh
05	Bangla Cluster Connect MSME Cluster Commerce & Proof-of-Origin Network	Middlemen, brand dilution and siloed ONDC pilots across 570 clusters	Siloed pilots	₹2.55–4.15 cr	₹50–85 lakh
06	Sabuj Shilpa Green Industry DMRV & Carbon-Readiness Platform	CCTS carbon targets and EU CBAM threatening jobs in heavy industry and exports	Not started at state level; our prototype is ready	₹1.25–2.0 cr	₹45–75 lakh
Whole portfolio				₹11.35–18.25 cr	₹3.3–5.55 cr

For scale: the whole portfolio costs about 0.2–0.3% of the ₹6,000 crore (incentive pool plus cluster programme) it would help the state spend well.

→ HOW THE SIX FIT TOGETHER

One investor, one ID, one journey. Nivesh Setu (01) tracks the investor from MoU to production. Shilpa Bhumi (02) finds and clears the land. Silpa Sathi 2.0 (03) issues approvals on a clock. Karma Praman (04) pays incentives against verified jobs. Bangla Cluster Connect (05) carries the same approach to MSME clusters, and Sabuj Shilpa (06) keeps heavy industry and exporters carbon-compliant. Each can be built on its own, but they share a common investor record, sign-on and dashboard, so every one added makes the others more useful.

RECOMMENDED STARTING POINTS

- **04 Karma Praman:** the most time-critical. The ₹5,000 crore jobs-linked policy is expected around October 2026, and building the verification engine alongside the policy means claims can be checked from the first payout.
- **06 Sabuj Shilpa:** fastest to show. Our DMRV prototype already works end to end, so a 20-unit pilot can go live in 8–10 weeks.
- **01 Nivesh Setu:** the shared backbone. A war-room for the top 100 projects gives the CM and Minister immediate visibility and links all the later modules.

Nivesh Setu

Investor Lifecycle & MoU-to-Ground Tracker

Every rupee promised to Bengal, tracked until it becomes a factory and a payroll.

OWNER & PARTNER DEPARTMENTS

ICE Department / WBIDC (nodal); CM's Office review cell; Land, Finance and Police (grievance helplines)

STATUS TODAY

Not started as an integrated system. Summit MoUs and the 33 newly cleared investments are tracked in separate lists; the police-run helplines (155334 / 155335) are not linked to investor files.

TIMELINE

Pilot 10–12 weeks · Full rollout 6–8 months

INDICATIVE COST

One-time ₹1.15–1.9 cr · Recurring ₹35–60 lakh / yr

1 PROBLEM STATEMENT

- From 2015 to 2019 Bengal's business summits announced ₹2.2–2.85 lakh crore of MoUs every year, but how many were actually built was never published. Critics and one central estimate put the conversion rate in single digits.
- The new government has cleared 33 fresh investments and promises time-bound approvals and investor handholding. But no single system shows, project by project, where each one is stuck and with whom.
- Complaints of extortion, syndicates and 'cut money' now go to police helplines (155334 / 155335). These tickets are not linked to the investor's project, so the ICE Department cannot see which investments are at risk.
- Data sits in Excel sheets across WBIDC, district offices and line departments. The CM's and Minister's review meetings depend on reports compiled by hand.

2 PROPOSED SOLUTION

A government-grade investor CRM that gives every investor one Investor ID and tracks each project through nine stages: Lead → MoU → Land identified → Land allotted → Clearances → Construction → Trial run → Commercial production → Jobs verified.

- Udyog Mitra desk:** Every project above ₹50 crore gets a named relationship officer. The officer works to service-level timers, and handovers are logged.
- Stage-gate tracker:** Automatic red / amber / green flags when any stage runs past its benchmark days, with escalation to the Secretary and then the Minister.
- Grievance bridge:** Helpline tickets that involve an investor show up in that project's file (with privacy controls) and are tracked to closure.
- War-room dashboard:** A live pipeline view for the CM and Minister, plus an auto-generated weekly review pack in Bengali and English.
- Public 'Investment Realised' board:** MoU value against money actually invested on the ground and verified jobs, by district and by sector.

UNIQUE DIFFERENTIATOR

Most states track MoUs. Few, if any, publicly track them all the way to verified jobs. Nivesh Setu closes that loop with EPFO/ESIC-verified job data from Project 04, so the state can show 'MoU → factory → payroll'. It is also the shared backbone for all six projects: one Investor ID across land, approvals, incentives and grievances.

3 APPROACH & METHODOLOGY

Phase	Duration	What we do	Output
0. Discovery	3 weeks	Map the current trackers, the 2026 MoU list and the 33 cleared projects. Agree the stages and service-level timelines with ICE and WBIDC.	Stage model and baseline
1. Pilot	8–10 weeks	War-room for the top 100 projects, the relationship-officer desk and the weekly review pack.	Live war-room
2. Rollout	4–5 months	All projects above ₹10 crore; grievance bridge; links to Silpa Sathi, Shilpa Bhumi and Karma Praman; public board.	State-wide tracker
3. Run	Ongoing	Operations and maintenance, quarterly improvements, and delay analytics (which step and which office delays most).	Monthly insights

Method: Agile, two-week sprints designed together with officers. Open-source stack (PostgreSQL, Keycloak single sign-on, React), hosted in the WB State Data Centre, with role-based access and full audit logs.

4 BENEFITS

For the Government

- Real-time view of the whole investment pipeline; review meetings take minutes instead of days of report-making.
- Escalations based on evidence; credible, publishable numbers for the next business summit.
- Early warning when an investor is facing harassment or delay.

For industry, MSMEs & citizens

- One contact, one ID and visible timelines for every investor.
- Complaints are tracked to closure instead of getting lost.
- Target KPIs (year 1): 100% of projects above ₹10 crore tracked; 30% cut in average stage delay; conversion rate published every quarter.

5 LIKELY COST RANGE (DEVELOPMENT + INFRASTRUCTURE)

Component	Low	High
Development: pilot (Phases 0–1)	₹25 lakh	₹40 lakh
Development: full rollout (Phase 2), incl. security audit	₹90 lakh	₹1.5 cr
One-time total	₹1.15 cr	₹1.9 cr
Infrastructure per year (hosting, storage, gateways, data)	₹15 lakh	₹25 lakh
Operations & maintenance per year (support, updates, helpdesk)	₹20 lakh	₹35 lakh
Recurring total per year	₹35 lakh	₹60 lakh

Infra covers hosting (State Data Centre or MeitY-empanelled cloud), SMS/e-mail gateway and backups. Figures exclude GST and are ±25% until scoped.

6 KEY RISKS & HOW WE HANDLE THEM

Departments don't keep data updated	Status is pulled automatically from Silpa Sathi and land systems where possible; data freshness is reviewed in the monthly meeting.
Investor data is commercially sensitive	Role-based access; the public board shows only aggregates.
Officers see it as surveillance	Positioned as support: each officer sees their own queue and can flag blockers upward.

Shilpa Bhumi

Industrial Land Intelligence & Closed-Factory Revival Grid

Know every industrial acre: who owns it, whether it is clean, and whether it is being used.

OWNER & PARTNER DEPARTMENTS

ICE; Land & Land Reforms (Banglarbhumii); WBIDC, WBIIDC, WBSIDCL; Public Enterprises & Industrial Reconstruction; District Magistrates

STATUS TODAY

Partially started (static data only). Land-availability lists are on WBIDC and ICE websites and about 570 MSME clusters were GIS-mapped earlier. A new land policy is being drafted. The analytics and revival layer has not started.

TIMELINE

Pilot 12 weeks (West Burdwan) · State rollout 6 months

INDICATIVE COST

One-time ₹2.15–3.55 cr · Recurring ₹75 lakh–1.3 cr / yr

1 PROBLEM STATEMENT

- Land is the number one constraint. The government has said the existing land bank cannot meet investor demand, and it is rewriting land policy after consulting industry (August 2026).
- Large tracts are locked inside closed units: about 2,100 acres in West Burdwan alone, including about 947 acres at Hindustan Cables, Rupnarayanpur. Several sites are under litigation, and District Magistrates prepare status reports by hand.
- The facts about a single plot are spread across Banglarbhumii (Record of Rights), Registration (encumbrance), WBIDC/WBIIDC allotments, the courts and the utilities. An investor cannot verify a plot without many office visits.
- Plots that were allotted in industrial parks but never used are not detected or taken back in any systematic way.
- The proposed policy to make land-givers stakeholders in new units needs a trustworthy record of who holds which stake.

2 PROPOSED SOLUTION

A single GIS record for every industrial parcel, built by combining all official sources, with satellite analytics on top and a step-by-step workflow for reviving closed-factory land.

- Parcel 360:** Click any plot to see its ownership (RoR), land class, encumbrance, court cases and allotment status, plus distance to power, water, road, rail and port (PM Gati Shakti layers).
- Idle-land radar:** Quarterly satellite change detection (ISRO Bhuvan plus high-resolution imagery where needed) that flags allotted plots with no activity for 24 months or more.
- Revival dossiers:** One file per closed factory: area, owner (central PSU, state or private), liabilities, litigation stage, next legal step and suitable sectors. Tracked on a revival board.
- Land-for-Equity registry:** Records each landowner's stake in the new industrial unit and their annual payouts. Certificates are issued through DigiLocker and updates sent by SMS.
- Investor Land Finder:** Filter plots by size, sector suitability and distance to port or highway, then book a site visit. The lead flows into Nivesh Setu.

UNIQUE DIFFERENTIATOR

Today's lists show what land is available. Shilpa Bhumi shows whether it is genuinely clean, whether allotted land is actually being used, and what it takes to unlock closed-factory land. The Land-for-Equity registry is a first-of-its-kind tool for the government's new consent-based land model: it makes the promise to landowners verifiable, which lowers the risk of land agitation.

3 APPROACH & METHODOLOGY

Phase	Duration	What we do	Output
0. Data compact	4 weeks	Data-sharing agreement across Land & Land Reforms, WBIDC, WBIIDC and Registration; inventory of existing map layers.	Signed data compact
1. Pilot	10–12 weeks	West Burdwan: 7 closed-factory sites and 2 WBIDC parks; drone survey of priority sites; revival dossiers; idle-plot scan.	District revival board
2. Rollout	5–6 months	All state industrial parks and land-bank parcels; public Land Finder; Land-for-Equity registry once the policy is notified.	State land grid

Phase	Duration	What we do	Output
3. Run	Ongoing	Quarterly satellite refresh, monthly data sync, dashboards for District Magistrates.	Quarterly idle-land report

Method: Open GIS stack (PostGIS, GeoServer, OpenLayers) on OGC standards. Integration with Banglarbhumii through an API or scheduled data transfer. Drone surveys to SVAMITVA-grade accuracy; the Law Department vets the dossier templates.

4 BENEFITS

For the Government

- Faster unlocking of idle and closed-factory land. Target: 1,000+ acres brought to 'investor-ready' status in year 1.
- Allotted-but-unused plots detected and reclaimed with evidence.
- An auditable, litigation-aware record for allotment decisions (useful for CAG and vigilance).

For industry, MSMEs & citizens

- Investors can verify a plot online in minutes instead of weeks.
- Fewer post-allotment disputes and delays.
- Landowners see their stake and payouts, which builds trust in the new land model.

5 LIKELY COST RANGE (DEVELOPMENT + INFRASTRUCTURE)

Component	Low	High
Development: pilot (Phases 0–1)	₹35 lakh	₹55 lakh
Development: full rollout (Phase 2), incl. security audit	₹1.8 cr	₹3.0 cr
One-time total	₹2.15 cr	₹3.55 cr
Infrastructure per year (hosting, storage, gateways, data)	₹40 lakh	₹75 lakh
Operations & maintenance per year (support, updates, helpdesk)	₹35 lakh	₹55 lakh
Recurring total per year	₹75 lakh	₹1.3 cr

Pilot includes a drone survey of priority sites (₹10–15 lakh). Infra includes hosting and commercial high-resolution satellite imagery. Figures exclude GST and are ±25% until scoped.

6 KEY RISKS & HOW WE HANDLE THEM

Reluctance to share data across departments	A data compact at Chief Secretary level; start read-only, with no change to the source systems.
Litigation-sensitive information made public	The public view shows only verified-clean plots; dossiers stay internal.
Satellite false positives	Field verification by the DM's team before any notice to reclaim land.

Silpa Sathi 2.0

Smart Clearance Engine: Rules-as-Code + Deemed-Approval Clock

Approvals in days, not months, with every decision explainable and every officer protected.

OWNER & PARTNER DEPARTMENTS

ICE / WBIDC (Silpa Sathi owner); Labour; Fire & Emergency; WBPCB; Urban Development & Municipal Affairs; Panchayat; Power; IT & Electronics / Webel Technology Ltd

STATUS TODAY

Partially started: overhaul underway. The portal is live, with kiosks at 23 district Industry Facilitation Centres. An overhaul announced in 2026 brings labour, fire and pollution clearances onto one form with payments and tracking. 'Deemed approval' is under debate. Rule-based automation is only at the prototype stage.

TIMELINE

Pilot 12 weeks (10 services) · Rollout 6 months (40+ services)

INDICATIVE COST

One-time ₹2.45–3.9 cr · Recurring ₹75 lakh–1.2 cr / yr

1 PROBLEM STATEMENT

- Even after the overhaul, most approvals still move from desk to desk. The portal collects documents but does not check them.
- Officers are rightly cautious. An automated decision without legal cover exposes them to vigilance and court scrutiny, which is why 'AI approvals' have stalled.
- New commitment: investors above ₹100 crore will not need to approach municipalities or panchayats, because the state will provide those clearances. The fast lane to deliver this does not exist yet.
- Deemed approval, if introduced, needs an automatic and tamper-proof clock. Without one it cannot be enforced.
- Investors enter the same data twice: once on the National Single Window System (NSWS) and again on state forms.

2 PROPOSED SOLUTION

A clearance engine that plugs into Silpa Sathi and does not replace it. It turns the rules into checkable code and puts every file on a clock.

- **Rules-as-Code library:** The checkable conditions of each Act and rule (setbacks, plot size, category, distance norms, fees) written as versioned machine rules that the Law Department has vetted. Applications are checked in seconds and gaps flagged to the applicant at once.
- **Assistive, not autonomous, AI:** The engine drafts a recommendation note that cites the rules. The officer approves in one click or records a reason to differ. Accountability stays with the officer, and the effort drops by 70–80%.
- **Statutory clock:** Each service runs on its notified timeline, with escalations at 50%, 80% and 100% of the time allowed. Where policy allows, a digitally signed deemed approval is issued automatically.
- **Mega-Project Fast Lane:** For projects of ₹100 crore or more: one state-level joint inspection and one consolidated approval packet covering the municipal and panchayat functions.
- **NSWS / CAF sync:** Enter data once at the national level and have it auto-filled on state forms; documents pulled from DigiLocker.

UNIQUE DIFFERENTIATOR

Rules-as-code with officer-assistive recommendations is not yet in use in West Bengal and is rare nationally. It turns 'deemed approval' from a slogan into an enforceable system. It also answers the IAS concern about legal accountability: every recommendation cites the rule, and every officer decision is recorded.

3 APPROACH & METHODOLOGY

Phase	Duration	What we do	Output
0. Baseline	4 weeks	Pick the 10 highest-volume services (factory licence, fire NOC, consent to establish / operate for green and orange categories, trade licence, park building plans). Run a time-and-motion baseline.	Service rule sheets
1. Pilot	12 weeks	Code the rules; Law Department vetting; shadow mode (the engine runs alongside officers and outcomes are compared, target ≥95% agreement) before go-live.	10 services automated
2. Rollout	5–6 months	40+ services, the statutory clock, the fast lane and NSWS sync.	Smart single window

Phase	Duration	What we do	Output
3. Run	Ongoing	A small 'rules desk' updates the code whenever a notification changes; delay analytics.	Always-current rules

Method: Open rules engine (OpenFisca/Drools class), eSign and digital signature certificates, and CERT-In and STQC audits before go-live. Bilingual interface. Shadow mode is the key to officer confidence.

4 BENEFITS

For the Government

- Target: 50–70% shorter approval time for covered services, and half as many rounds of deficiency notices.
- Officers protected by rule citations and a full audit trail.
- A direct lift to the state's Business Reform Action Plan (BRAP) ranking.

For industry, MSMEs & citizens

- Predictable timelines and instant feedback on gaps in an application.
- The ₹100 crore fast-lane promise becomes real.
- No duplicate data entry between national and state portals.

5 LIKELY COST RANGE (DEVELOPMENT + INFRASTRUCTURE)

Component	Low	High
Development: pilot (Phases 0–1)	₹45 lakh	₹70 lakh
Development: full rollout (Phase 2), incl. security audit	₹2.0 cr	₹3.2 cr
One-time total	₹2.45 cr	₹3.9 cr
Infrastructure per year (hosting, storage, gateways, data)	₹30 lakh	₹50 lakh
Operations & maintenance per year (support, updates, helpdesk)	₹45 lakh	₹70 lakh
Recurring total per year	₹75 lakh	₹1.2 cr

O&M includes a 2-person rules desk that keeps the coded rules in line with new notifications. The rollout estimate includes the CERT-In security audit. Figures exclude GST and are ±25% until scoped.

6 KEY RISKS & HOW WE HANDLE THEM

Legal accountability for automated decisions	Assistive mode, Law Department vetting, and shadow mode before go-live.
Rules change frequently	A dedicated rules desk; every rule versioned and dated.
Resistance from line departments	Start with services ICE controls; publish the time saved to build demand.

Karma Praman

Jobs-Linked Incentive Verification & Disbursal Engine

Pay incentives only for real jobs: verified, automatic and leak-proof.

OWNER & PARTNER DEPARTMENTS

ICE (incentive owner); Finance (IFMS); Labour. Data partners: EPFO, ESIC, GSTN / State GST, WBSEDCL

STATUS TODAY

Not started: greenfield. The new industrial policy, with a ₹5,000 crore jobs-linked incentive pool in the FY27 budget, is in its final stage (expected around October 2026). No verification system exists yet.

TIMELINE

Pilot 12 weeks · Rollout 5 months: ready before the first claims fall due

INDICATIVE COST

One-time ₹1.8–2.75 cr · Recurring ₹50–85 lakh / yr

1 PROBLEM STATEMENT

- The new policy links incentives to jobs created rather than money invested. The idea is right, but 'jobs' are easy to inflate: ghost employees, short-term hires dropped after the count, workers shuffled between group companies.
- The previous incentive regime ended in dispute. A 2025 law withdrew incentives dating back to 1993, and industry says more than ₹20,000 crore of claims are outstanding. Trust has to be rebuilt through transparency.
- Checking payroll documents by hand for thousands of units is slow and leaves room for discretion, which is exactly what investors complain about.
- EPFO is now considering sharing real-time data with states (August 2026). This is the window to build the system right from the start.

2 PROPOSED SOLUTION

A policy-as-code engine that calculates, verifies and pays out incentives from trusted government data.

- Policy-as-code + simulator:** Each eligibility and payout rule is coded once and version-controlled. Finance can simulate the budget impact of a rule before it is notified.
- Jobs verification:** Monthly EPFO/ESIC data, with the unit's consent: new UANs, wage bands, share of Bengal residents, women and SC/ST shares where the policy rewards them. Retention checks at 12 and 24 months.
- Activity cross-checks:** GST returns and electricity consumption confirm that the unit is really operating.
- Anomaly analytics:** Flags the same UAN appearing across group entities, hiring spikes just before a claim, and wage outliers.
- Auto-disbursal + legacy claims:** Approved claims go to IFMS for direct transfer to the unit's bank account, with claw-back if jobs fall. A separate module digitises old claims and classifies them as valid, disputed or time-barred.

UNIQUE DIFFERENTIATOR

This is a state incentive scheme in which the money follows payroll data automatically. It works alongside the Centre's ELI / PM-VBRY scheme, which already relies on EPFO data. It can be designed while the policy is still being written, so the rules are ready for the system on day one, and it gives the government a fact base for settling legacy dues.

3 APPROACH & METHODOLOGY

Phase	Duration	What we do	Output
0. Co-draft	3–4 weeks	Work with the policy drafting team to turn the draft into rule tables. Put data-sharing agreements in place (EPFO, ESIC, GST).	Coded policy + simulator
1. Pilot	10–12 weeks	Claims portal plus file-based EPFO verification for the first 50 applicant units.	First verified claims
2. Rollout	4–5 months	API integrations, anomaly engine, IFMS direct transfer, legacy-claims module and public dashboard.	End-to-end engine
3. Run	Ongoing	Monthly verification runs; annual policy updates; reports ready for CAG audit.	Monthly payout cycle

Method: Privacy by design under the DPDP Act 2023 (consent, purpose limitation, masked Aadhaar and UAN). A ledger that is ready for audit, and every payout traceable to the rule version and data used.

4 BENEFITS

For the Government

- Every rupee tied to a verified job. Preventing even 1% leakage on ₹5,000 crore saves ₹50 crore, about 20 times what the system costs.
- Officers freed from manual document checks; decisions that stand up to scrutiny.
- A fact base for settling legacy incentive dues fairly.

For industry, MSMEs & citizens

- Faster, predictable payouts. Target: within 30 days of the claim period closing.
- No discretion or middlemen, so investor confidence rises.
- Public dashboard: incentive paid against jobs verified, by district and sector.

5 LIKELY COST RANGE (DEVELOPMENT + INFRASTRUCTURE)

Component	Low	High
Development: pilot (Phases 0–1)	₹30 lakh	₹45 lakh
Development: full rollout (Phase 2), incl. security audit	₹1.5 cr	₹2.3 cr
One-time total	₹1.8 cr	₹2.75 cr
Infrastructure per year (hosting, storage, gateways, data)	₹20 lakh	₹40 lakh
Operations & maintenance per year (support, updates, helpdesk)	₹30 lakh	₹45 lakh
Recurring total per year	₹50 lakh	₹85 lakh

Excludes any fees charged by EPFO, ESIC or GSTN for data access (normally nil for government use). Figures exclude GST and are ±25% until scoped.

6 KEY RISKS & HOW WE HANDLE THEM

Delays in API access to EPFO / ESIC	Start by verifying the ECR returns units upload; switch to APIs when they are available.
Privacy of worker data	DPDP-compliant consent; masked identifiers; purpose-limited use.
Policy amended after launch	Policy-as-code with versioning; the simulator tests each change before notification.

Bangla Cluster Connect

MSME Cluster Commerce & Proof-of-Origin Network

From Bengal's looms, tea gardens and tanneries straight to buyers, with proof of origin.

OWNER & PARTNER DEPARTMENTS MSME & Textiles; ICE; WBSIDCL; Biswa Bangla. Partners: Tea Board, Jute Commissioner, ONDC, SIDBI, GeM	STATUS TODAY Siloed pilots. About 570 clusters mapped. A ₹1,000 crore Legacy Industrial Cluster Rejuvenation programme (jute in Howrah and Hooghly, tea in Siliguri and Darjeeling) has been announced. ONDC onboarding is scattered across departments, and there is no state-level cluster platform.
TIMELINE Pilot 12 weeks (2 clusters) · Scale-up 6 months (20 clusters)	INDICATIVE COST One-time ₹2.55–4.15 cr · Recurring ₹50–85 lakh / yr

1 PROBLEM STATEMENT

- About 570 MSME clusters (jute, handloom, leather, foundry, garments, zari, tea) sell mostly through middlemen, so producers keep thin margins.
- ONDC pilots are run separately by different departments. Nobody owns cluster-level onboarding, catalogues, logistics and payments.
- The brand is being diluted: industry estimates suggest nearly 40% of exported 'Darjeeling' tea is blended or adulterated, and Bengal handlooms face powerloom copies. The GI premium leaks away.
- Leather (Bantala), garments (Metiabruz) and gems & jewellery have no dedicated intervention in the budget.
- Export buyers increasingly demand digital traceability and ESG evidence that micro units cannot produce on their own.

2 PROPOSED SOLUTION

A state-backed cluster commerce layer and traceability backbone. It is digital public infrastructure, not another marketplace.

- ONDC Seller Gateway:** Bulk onboarding of cluster units through an ONDC seller-side partner. Voice-first catalogue creation in Bengali (photo → listing), pricing help, and logistics and payments through ONDC.
- Cluster Digital Passport:** A QR or NFC tag on each lot carrying the origin garden or loom, GI registration, artisan ID and certification. Buyers scan to verify. Tamper-evident ledger.
- B2B demand board:** Bulk buyers (hotels, exporters, government buyers on GeM) post requirements and cluster SPVs bid together.
- Credit link:** Sales history feeds lending by SIDBI and banks, and invoice discounting on TReDS.
- Programme dashboard:** For the ₹1,000 crore programme: units onboarded, sales, prices and artisan earnings, tracking outcomes rather than just spend.

UNIQUE DIFFERENTIATOR

This plugs Bengal's clusters into ONDC and GeM as one state network and proves authenticity lot by lot. It gives the ₹1,000 crore rejuvenation programme measurable outcomes (sales and income) instead of spending figures alone, and it can pay for itself through a small transaction fee.

3 APPROACH & METHODOLOGY

Phase	Duration	What we do	Output
0. Select	4 weeks	Choose 2 pilot clusters (e.g. Hooghly jute products and Santipur–Phulia handloom or a Darjeeling estate group); sign an ONDC seller-side partner.	Cluster plans
1. Pilot	12 weeks	Onboard 500 units through cluster facilitators and IFCs; digital passports for 2 GI products.	First sales on ONDC
2. Scale-up	6 months	20 clusters and 10,000 units; B2B board; credit link; programme dashboard.	State cluster network
3. Sustain	Ongoing	Self-financing through a 0.5–1% transaction fee or a subscription for cluster SPVs.	Low cost to state

Method: Field first: cluster facilitators and the Industry Facilitation Centres do the onboarding. Vernacular, voice-led screens; open network standards (Beckn / ONDC); data shared only with the unit's consent.

4 BENEFITS

For the Government

- Measurable outcomes for the ₹1,000 crore cluster programme.
- The GI premium protected and Bengal's export credibility raised.
- Formal sales records bring micro units into the tax and credit system.

For industry, MSMEs & citizens

- Target: 15–25% better prices for onboarded units, because less is lost to middlemen.
- Direct access to national buyers and to government procurement on GeM.
- Access to credit based on verified sales.

5 LIKELY COST RANGE (DEVELOPMENT + INFRASTRUCTURE)

Component	Low	High
Development: pilot (Phases 0–1)	₹35 lakh	₹55 lakh
Development: full rollout (Phase 2), incl. security audit	₹1.6 cr	₹2.6 cr
Field onboarding (cluster facilitators, training)	₹60 lakh	₹1.0 cr
One-time total	₹2.55 cr	₹4.15 cr
Infrastructure per year (hosting, storage, gateways, data)	₹20 lakh	₹35 lakh
Operations & maintenance per year (support, updates, helpdesk)	₹30 lakh	₹50 lakh
Recurring total per year	₹50 lakh	₹85 lakh

Field onboarding means cluster facilitators, photography and training. QR/NFC tags cost ₹1–5 each, borne by the producer or the cluster programme. Figures exclude GST and are ±25% until scoped.

6 KEY RISKS & HOW WE HANDLE THEM

Low digital literacy	Cluster facilitators, voice-first Bengali screens, and assisted listing at IFCs.
High logistics cost for small orders	Cluster aggregation points and shared shipments.
Platform dies after the pilot	A transaction-fee or SPV-subscription model is designed in from day one.

Sabuj Shilpa

Green Industry DMRV & Carbon-Readiness Platform

Keep Bengal's industry compliant and export-ready, and let it earn from going green.

OWNER & PARTNER DEPARTMENTS

ICE; Non-Conventional & Renewable Energy Sources (same Minister); Environment / WBPCB. Partners: BEE, industry associations, accredited carbon verifiers

STATUS TODAY

Not started at state level; our prototype is ready. The national Indian Carbon Market portal went live in March 2026. We are not aware of any state that offers its industry a service to help it comply. Our working DMRV prototype already covers onboarding → data → verification → registry.

TIMELINE

Pilot 8–10 weeks (20 units) · Rollout 5 months

INDICATIVE COST

One-time ₹1.25–2.0 cr · Recurring ₹45–75 lakh / yr

1 PROBLEM STATEMENT

- India's Carbon Credit Trading Scheme (CCTS) is now binding. About 490 obligated entities in 7 sectors (aluminium, cement, chlor-alkali, pulp & paper, refinery, petrochemicals, textiles) must cut emission intensity in FY26 and FY27. Iron & steel and fertiliser targets come next, and they fall directly on the Durgapur–Asansol–Haldia belt.
- The EU's Carbon Border Adjustment Mechanism (CBAM) entered its paid phase in January 2026. Bengal's steel, aluminium and engineering exporters, including MSME suppliers, need verified product-level emission data or they lose EU buyers.
- Most units, and especially MSMEs and secondary steel and sponge-iron plants, lack the capacity to measure, report and verify (MRV) their emissions. Consultants are costly and the data is collected by hand.
- WBPCB already receives online emission data (OCEMS) from red-category units, but that data is not used for economic or green-investment decisions.

2 PROPOSED SOLUTION

A state-sponsored digital MRV (DMRV) service for industry, built on our existing working prototype.

- Facility registry + auto-ingest:** Registry in CCTS format; energy bills, fuel purchases, production data and WBPCB OCEMS feeds loaded automatically.
- CCTS compliance engine:** Emission-intensity calculation to the official methodology, alerts on the gap to target, and ready-made packs for accredited verifiers and BEE.
- CBAM Export Desk:** Calculators for the emissions embedded in each product, and reports in the EU's format, for MSME exporters.
- Green incentive link:** Verified reductions (rooftop solar, efficiency upgrades) qualify for state green top-ups under the new industrial policy.
- State Green Industry Dashboard + aggregation:** A sector and district view for the Renewable Energy portfolio, and pooling of MSME efficiency gains to earn carbon credits under the offset mechanism.

UNIQUE DIFFERENTIATOR

The state provides this to its industry as a service, instead of leaving each unit to face Delhi and Brussels alone. We already have a working prototype of the full workflow (entity onboarding → data quality → calculation lineage → third-party verification → regulator approval → registry), so the pilot starts in weeks, not months.

3 APPROACH & METHODOLOGY

Phase	Duration	What we do	Output
0. Mobilise	3 weeks	List the obligated entities based in Bengal and the top 50 EU exporters; agree the OCEMS data feed with WBPCB; workshops in Durgapur and Haldia.	Pilot cohort
1. Pilot	8–10 weeks	Configure the prototype for 20 units (large obligated entities plus MSME exporters); first CCTS and CBAM reports.	First compliance packs
2. Rollout	5 months	All willing obligated entities plus 500 MSMEs; aggregation; dashboard; incentive link.	State DMRV service
3. Sustain	Ongoing	Large units pay a modest subscription; the state funds access for MSMEs.	Self-funding core

Method: Methodologies aligned with BEE and CCTS, a full audit trail and lineage for every calculated number, and optional IoT energy meters for high-emitting units. Data stays within India.

4 BENEFITS

For the Government

- The state's first industrial carbon map, useful for climate finance and for promoting green investment.
- Keeps export jobs in the Durgapur–Asansol and Howrah engineering belts.
- Low long-term cost to the state, because subscriptions and aggregation fees pay for the service.

For industry, MSMEs & citizens

- Obligated units avoid penalties and can earn and sell Carbon Credit Certificates.
- MSMEs keep their EU orders under CBAM.
- Compliance at a fraction of what a consultant would charge.

5 LIKELY COST RANGE (DEVELOPMENT + INFRASTRUCTURE)

Component	Low	High
Development: pilot (Phases 0–1)	₹25 lakh	₹40 lakh
Development: full rollout (Phase 2), incl. security audit	₹1.0 cr	₹1.6 cr
One-time total	₹1.25 cr	₹2.0 cr
Infrastructure per year (hosting, storage, gateways, data)	₹20 lakh	₹35 lakh
Operations & maintenance per year (support, updates, helpdesk)	₹25 lakh	₹40 lakh
Recurring total per year	₹45 lakh	₹75 lakh

Lower than a greenfield build because the core prototype exists. Optional IoT energy meters cost ₹0.5–2 lakh per unit, borne by the industry or subsidised. Figures exclude GST and are ±25% until scoped.

6 KEY RISKS & HOW WE HANDLE THEM

Steel and fertiliser targets not yet notified	Configurable methodology library; build to the draft norms now.
Units hesitant to share emission data	Voluntary and confidential, and it gives value first (CBAM reports buyers demand).
BEE or EU methodology changes	Versioned calculations with full lineage; re-run history on demand.

How we will work with the Department

Start small, prove value	A 6–8 week proof of concept on one initiative, using the Department's own data in its own environment, at no cost or a nominal one. Scale up only after the results are visible.
Clean, transparent procurement	Engagement through GeM, Webel Technology Ltd (the state's nodal e-governance agency) or an empanelled system-integrator partner. Rollouts through open tender, where we compete on merit.
Data stays with the state	Hosting in the WB State Data Centre or a MeitY-empanelled Indian cloud. Compliant with the DPDP Act 2023, with consent-based data sharing.
Security first	A CERT-In-empanelled security audit before every go-live, with STQC certification where required. Role-based access and tamper-evident audit logs.
No vendor lock-in	Open-source stack, source code in escrow, full documentation, and training for state officers. The state owns its configurations and data.
Bengali-first	Screens for citizens and MSMEs in Bengali, Hindi and English, with voice-led flows for artisans and micro units.
Outcome-linked contracts	Every engagement carries measurable KPIs (approval days, jobs verified, acres unlocked) and monthly public-facing reports.
Build in Bengal	Delivery team based in Kolkata, with internships and hiring from state engineering colleges and polytechnics.

i COST ASSUMPTIONS

- Blended delivery cost of ₹1.4–2.0 lakh per person-month for a Kolkata-based team (product, design, engineering, QA, project management).
- Pilot = Phases 0–1; rollout = Phase 2. Operations and maintenance are typically 15–20% of build cost per year and include the helpdesk, bug fixes, minor enhancements and rule updates.
- Infrastructure assumes a MeitY-empanelled cloud. If hosted in the State Data Centre, part of this cost is absorbed by the state.
- All amounts exclude GST and are indicative (±25%) until scoped jointly with the Department.

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Status descriptions are based on public reports up to September 2026 and should be confirmed with the Department before formal submission.